

EASYRIDE (Car Booking System)

- **Honey**
- **25BSCS007 (CS-A)**
- **Programming Fundamentals Project**

EASYRIDE (Car Booking System) Source Code

```
#include <iostream>
#include <string>
using namespace std;
struct Owner {
    string name, cnic, city, phone, email;
    string password;
};
struct Customer {
    string name, cnic, city, phone;
    string password;};
struct Car {
    string name;
    int rent;
    int category; // 1=SUV, 2=Hatchback, 3=Sports
    bool available;};
Car cars[100];
int carCount = 0;
void addCar() {
```

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    Car c;
    cin.ignore();
    cout << "Enter Car Name: ";
    getline(cin, c.name);
    cout << "Enter Rent per Day: ";
    cin >> c.rent;
    cout << "Enter Category (1=SUV, 2=Hatchback, 3=Sports): ";
    cin >> c.category;
    c.available = true;
    cars[carCount++] = c;
    cout << "Car added successfully!\n";
}

void removeCar() {
    int index;
    cout << "Enter car number to remove: ";
    cin >> index;
    index--;
    if (index >= 0 && index < carCount) {
        for (int i = index; i < carCount - 1; i++) {
            cars[i] = cars[i + 1]; }
        carCount--;
        cout << "Car removed successfully!\n";
    } else {
        cout << "Invalid car number!\n";
    }
}

void updateRent() {
    int index;

```

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cout << "Enter car number to update rent: ";
cin >> index;
index--;
if (index >= 0 && index < carCount) {
    cout << "Enter new rent: ";
    cin >> cars[index].rent;
    cout << "Rent updated successfully!\n";
} else {
    cout << "Invalid car number!\n";
}
}

void createOwnerAccount(Owner &o) {
    cin.ignore();
    cout << "\n--- Create Owner Account ---\n";
    cout << "Name: ";
    getline(cin, o.name);
    cout << "CNIC: ";
    getline(cin, o.cnic);
    cout << "City: ";
    getline(cin, o.city);
    cout << "Phone: ";
    getline(cin, o.phone);
    cout << "Email: ";
    getline(cin, o.email);
    cout << "Set Password: ";
    getline(cin, o.password);
    cout << "\nOwner account created successfully!\n";
}

void ownerMenu(Owner o, bool created) {

```

```

if (!created) {
cout << "No owner account found.\n";
return; }

string cnic, pass;
cin.ignore();
cout << "Enter CNIC: ";
getline(cin, cnic);
cout << "Enter Password: ";
getline(cin, pass);
if (cnic != o.cnic || pass != o.password) {
cout << "Invalid login credentials!\n";
return; }
int choice;
do {cout << "\n--- Owner Menu ---\n";
cout << "1. Add Car\n2. Remove Car\n3. Update Rent\n4. Logout\n";
cin >> choice;
if (choice == 1) addCar();
else if (choice == 2) removeCar();
else if (choice == 3) updateRent();
} while (choice != 4);}

void customerMenu() {
int choice;
do {
cout << "\n--- Customer Menu ---\n";
cout << "1. View Cars\n2. Book Car\n3. Logout\n";
cin >> choice;
if (choice == 1) {

```

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for (int i = 0; i < carCount; i++) {
    cout << i + 1 << ". " << cars[i].name
        << " | Rent: " << cars[i].rent
        << " | " << (cars[i].available ? "Available" : "Booked")
        << endl;
    }
else if (choice == 2) {
    int num, days;
    cout << "Enter car number: ";
    cin >> num;
    num--;
    if (num >= 0 && num < carCount && cars[num].available) {
        cout << "Enter days: ";
        cin >> days;
        int total = days * cars[num].rent;
        cars[num].available = false;
        cout << "Car booked successfully!\n";
        cout << "Total Rent: " << total << endl;
    } else {
        cout << "Car not available!\n";
    }
} while (choice != 3);}

void createCustomerAccount(Customer &c) {
    cin.ignore();
    cout << "\n--- Create Customer Account ---\n";
    cout << "Name: ";
    getline(cin, c.name);
}

```

```

cout << "CNIC: ";
getline(cin, c.cnic);
cout << "City: ";
getline(cin, c.city);
cout << "Phone: ";
getline(cin, c.phone);
cout << "Set Password: ";
getline(cin, c.password);
cout << "\nCustomer account created successfully!\n";
}

void customerLogin(Customer c, bool created) {
    if (!created) {
        cout << "No customer account found.\n";
        return; }
    string cnic, pass;
    cin.ignore();
    cout << "Enter CNIC: ";
    getline(cin, cnic);
    cout << "Enter Password: ";
    getline(cin, pass);
    if (cnic != c.cnic || pass != c.password) {
        cout << "Invalid login credentials!\n";
        return;}
    cout << "Login successful!\n";
    customerMenu();}

int main() { Owner owner;

```

```

Customer customer;

bool ownerCreated = false;

bool customerCreated = false;

int choice;

do {
    cout << "\n=== EASYRIDE ===\n";
    cout << "1. Owner\n2. Customer\n3. Exit\n";
    cin >> choice;
    if (choice == 1) {
        int ch;
        do {
            cout << "1. Create Account\n2. Login\n3. Back\n";
            cin >> ch;

            if (ch == 1) {
                createOwnerAccount(owner);
                ownerCreated = true; }
            else if (ch == 2) {
                ownerMenu(owner, ownerCreated);}
        } while (ch != 3); }
    else if (choice == 2) {
        int ch;
        do {
            cout << "1. Create Account\n2. Login\n3. Back\n";
            cin >> ch;
            if (ch == 1) {
                createCustomerAccount(customer);

```

```

customerCreated = true;}

else if (ch == 2) {

customerLogin(customer, customerCreated);}

} while (ch != 3);}

} while (choice != 3);

cout << "Thank you for using EasyRide!\n";

return 0;}

```

output:

```

9:16 AM
Compile Result

=== EASYRIDE ===
1. Owner
2. Customer
3. Exit
1
1. Create Account
2. Login
3. Back
1
--- Create Owner Account ---
Name: honey
CNIC: 1000
City: hyd
Phone: 030000
Email:
Set Password: 1234

Owner account created successfully!
1. Create Account
2. Login
3. Back
2
Enter CNIC: 1000
Enter Password: 1234

--- Owner Menu ---
1. Add Car
2. Remove Car
3. Update Rent
4. Logout
1
Enter Car Name:

```

```

--- Owner Menu ---
1. Add Car
2. Remove Car
3. Update Rent
4. Logout
2
Enter car number to remove: 1
Car removed successfully!

--- Owner Menu ---
1. Add Car
2. Remove Car
3. Update Rent
4. Logout
4
1. Create Account
2. Login
3. Back
3
=== EASYRIDE ===
1. Owner
2. Customer
3. Exit
2
1. Create Account
2. Login
3. Back
1
--- Create Customer Account ---
Name: maheen
CNIC: 2000
City: hyd
Phone: 0300
Set Password: 3456

Customer account created successfully!
1. Create Account
2. Login
3. Back

```